

Nondestructive Techniques for Fruit Ripeness Determination Using Machine Learning Classifiers

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Abstract

In this paper, we developed and evaluated non-destructive techniques to determine the ripeness state of local berry fruits. The ripeness was modeled as a binary classification problem (ripe versus unripe). We compared five Machine Learning (ML) methods: Decision Tree (DT), Quadratic Discriminant Analysis (QDA), Support Vector Machine (SVM), Ensemble Learning, and k -Nearest Neighbors (k -NN). Infrared (IR) thermal images of freshly plucked berries were captured at two distinct instances. The relative size of the higher-temperature inner cores and their corresponding average temperatures were recorded to serve as the primary predictors and features for each classification model.

Keywords: non-destructive testing, fruit ripeness, machine learning classifiers, MajorAxisLength, Area, Inner Core, infrared.

1 Introduction

Achieving uniform quality in both raw materials and final products is a primary objective within the food industry. For the fruit sector, the systematic determination of fruit ripeness under harvest and post-harvest conditions remains a core concern, as variability in maturity is perceived by consumers as a deficiency in product quality.

Most traditional methods used to assess fruit ripeness are destructive, limiting their scalability and real-time application. Some of these conventional techniques rely on measuring fruit firmness—which correlates directly with ripeness—using a penetrometer [1] or by evaluating impact force [2]. Other strategies involve quantifying chemical parameters that correlate with fruit maturity, such as pH, titratable acidity [3], soluble solids content (i.e., sugars) [4], and ethylene emissions [5]. Extrapolating these metrics typically mandates destroying the fruit sample alongside executing complex analytical protocols, such as gas-liquid chromatography for ethylene tracking and chemical titration for acidity mapping [6].

To overcome these constraints, non-destructive methodologies for determining fruit ripeness have recently been proposed. These prominent approaches include:

- (i) Nuclear Magnetic Resonance (NMR) [7] and Proton Magnetic Resonance (PMR) [8] to quantify internal soluble solids,
- (ii) Computer vision systems to evaluate external fruit skin color, which has demonstrated correlations with internal sugar concentration [9], and
- (iii) Acoustic methods to estimate internal fruit firmness [10].

However, technical and economic limitations persist across these frameworks. NMR and PMR systems rely heavily on expensive, laboratory-grade instrumentation. Acoustic methods face operational hurdles regarding how to reproducibly couple the acoustic impedances of the signal emitter and the fruit structure under test. Furthermore, external fruit color can be weakly or inconsistently correlated with true internal maturity.

An attractive, alternative strategy for classifying fruit ripeness involves deploying trained Machine Learning (ML) classifiers on computationally efficient hardware platforms. In this work, the optimized classification models successfully processed response vectors, yielding validation accuracies of 58.3% for the Complex Decision Tree (DT), 66.7% for the Quadratic Discriminant Analysis (QDA), 91.7% for the Quadratic Support Vector Machine (SVM), 83.3% for the Fine k -Nearest Neighbors (k -NN), and 91.7% when utilizing the Ensemble Bagged Trees architecture.

2 Methodology

2.1 Materials

The berries were freshly plucked from trees within the institute campus at the Indian Institute of Technology, Kharagpur. They were then gently brushed to remove dust or any other foreign particles that could interfere with the fruit's thermal signature. Two separate batches of berries were prepared for the experiments. The first batch, collected on 2nd March 2018, served as the training set, whereas the second batch, collected on 1st May 2018, was utilized as the test set to validate classification accuracy.

The Infrared (IR) spectrum thermal images were captured using a commercial IR camera featuring a temperature resolution of 0.1 K. The temperature distribution across various regions of the fruit was extracted directly from these thermal signatures, as illustrated in Fig. 1.

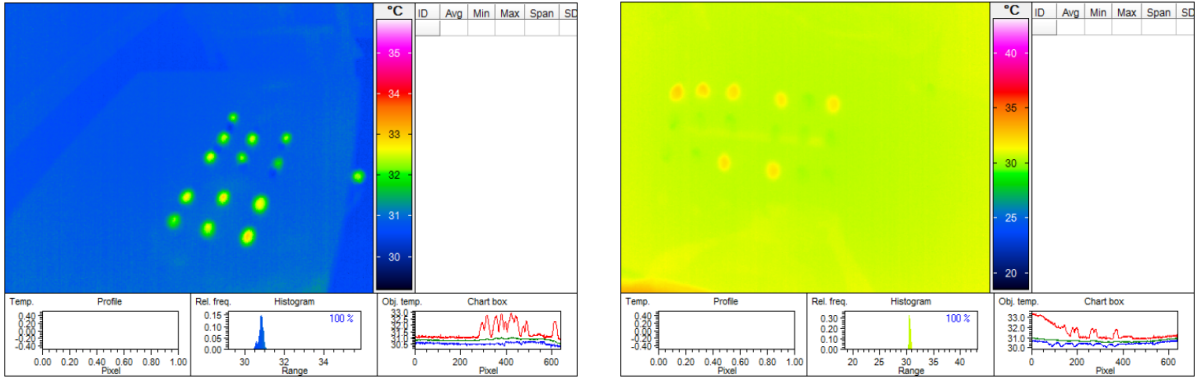


Figure 1: Infrared thermal images showing the training set batch (left) and the validation test set batch (right).

These raw thermal profiles were extensively processed and analyzed using MATLAB® (version R2017a). The primary computational tools and features employed in this workflow include:

- **Image Processing Toolbox:** This module provides reference-standard algorithms and workflow applications for image processing, analysis, visualization, and algorithm development. It was utilized to execute image segmentation, contrast enhancement, noise reduction, geometric transformations, and automated feature extraction.
- **Classification Learner App:** This interactive workspace trains predictive models to classify categorical data. It was leveraged to explore the extracted feature space, select optimal predictors, specify cross-validation schemes, and evaluate classifier performance metrics. Automated training routines were executed to sweep for the best-performing classification model types, including DT, discriminant analysis, SVM, logistic regression, k -NN, and ensemble classification architectures.

Supervised ML was performed by providing the ecosystem with a labeled calibration set containing known input observations and corresponding target classes (i.e., ripe versus unripe

labels). This reference framework was used to optimize a model capable of generating robust target predictions when exposed to novel boundary data. Once finalized, the optimized deployment models were exported to the global workspace for test verification.

2.2 Test Procedures

The data extraction and sampling process for the training and testing sets involved the following sequential steps:

- (i) Capturing IR thermal images of individual berry samples.
- (ii) Extracting three key structural and thermal features: average core temperature and relative core dimensions quantified by *Area* and *MajorAxisLength*.
- (iii) Computing the relative core size ratios according to the following metrics:
 - a) Relative Area = Area of berry inner core $\div$ Area of complete berry
 - b) Relative Axis Length = *MajorAxisLength* of berry inner core $\div$ *MajorAxisLength* of complete berry

The structural variations derived from these geometric operations are visualized in the processed binary masks shown in Fig. 2.

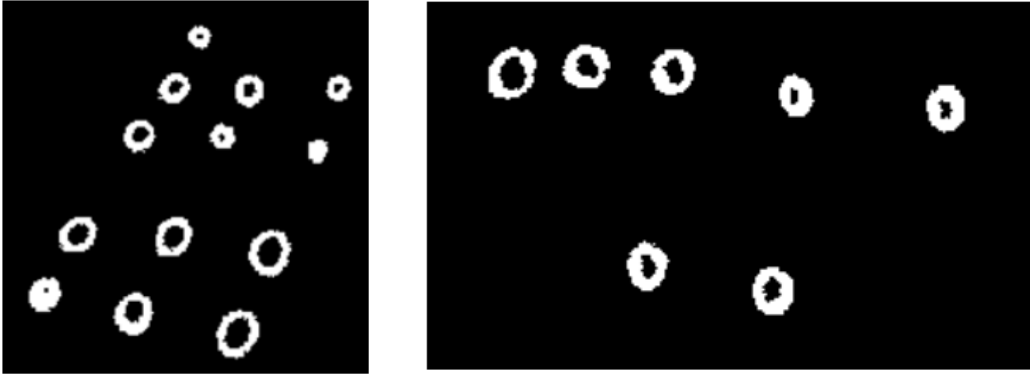


Figure 2: Processed structural segmentations showing the training batch masks (left) and the validation test batch masks (right).

- (iv) Training the supervised ML algorithms within the Classification Learner application utilizing the compiled feature matrix.
- (v) Evaluating and comparing the trained classifiers based on validation accuracy. The optimal configuration from each of the five core categories—DT, SVM, k -NN, discriminant analysis, and ensemble learning—was exported for subsequent performance validation on the independent test batch.

2.3 The Use of High-Temperature Inner Cores as a Class Reference

Thermal treatment protocols for fruits using hot water immersion, vapor heat, or forced hot air have been investigated extensively as non-chemical alternatives to methyl bromide fumigation for quarantine fruit disinfestation [11, 12, 13].

Varying degrees of efficacy have been reported using different standalone thermal treatments or when combined with cold storage or controlled atmosphere (CA) conditions [14, 15, 16]. Because target pests frequently reside deep within the geometric center of the fruit, thermal energy must be delivered effectively to the internal core region.

A primary limitation of conventional forced hot air or water immersion protocols is the slow rate of convective and conductive heat transfer, often requiring hours of total processing time [17]. Documented heating times for the exact center of a fruit to reach required insecticidal temperatures range from 23 min for small cherries up to 6 h for dense apples. The rate of thermal energy accumulation within the fruit core is significantly influenced by overall fruit volume, boundary heating medium temperature, and fluid velocity. Slow conductive heat propagation frequently causes localized core heating times to exceed 60 min [18].

Consequently, transient temperature gradients between the fruit surface and its internal core persist due to differences in material thermal diffusivities [18]. This thermal behavior exhibits a strong dependence on overall fruit size, heating medium temperature, fluid velocity, and the specific application method. This spatial temperature variance justifies using the high-temperature internal core footprint as a reliable feature for non-destructive ML sorting systems.

3 Data Analysis

3.1 Observations

The experimental data collected from both berry batches were structured around three core features: Area Ratio, *MajorAxisLength* Ratio, and Core Temperature. Table 1 outlines the feature profile extracted from the training set calibration batch.

Table 1: Extracted feature observations for the training calibration batch (captured on 2nd March 2018).

S.L. No.	Ripe State			Unripe State		
	Area Ratio	<i>MajorAxisLength</i>	Temp (°C)	Area Ratio	<i>MajorAxisLength</i>	Temp (°C)
1	0.022099	0.135933	32.1	0.344262	0.494994	32.6
2	0.474026	0.505642	32.6	0.232000	0.450670	32.4
3	0.338308	0.477238	32.5	0.077922	0.314828	32.2
4	0.479042	0.508862	32.7	0.072917	0.313707	32.2
5	0.626214	0.575305	32.8	0.236220	0.439575	32.5
6	0.592040	0.567788	32.6	0.000000	0.000000	31.9
7	-	-	-	0.144444	0.384934	32.2

To validate classification robustness, an independent test set was evaluated, with its structural parameters detailed in Table 2.

Table 2: Extracted feature observations for the independent validation test batch (captured on 1st May 2018).

S.L. No.	Ripe State			Unripe State		
	Area Ratio	<i>MajorAxisLength</i>	Temp* (°C)	Area Ratio	<i>MajorAxisLength</i>	Temp* (°C)
1	0.444795	0.692297	32.7	0.199275	0.507095	32.1
2	0.352941	0.626633	32.6	0.092308	0.403384	31.9
3	0.442798	0.566756	32.3	0.116935	0.430417	31.9
4	0.475472	0.579162	32.3	-	-	-

* Note: Test batch temperatures were increased by 0.2 °C to account for ambient environmental variations

3.2 Training of Supervised Machine Learning Models

1.1 ☆ Tree	Accuracy: 58.3%
Last change: Complex Tree	3/3 features
1.2 ☆ Tree	Accuracy: 58.3%
Last change: Medium Tree	3/3 features
1.3 ☆ Tree	Accuracy: 58.3%
Last change: Simple Tree	3/3 features
2.1 ☆ Linear Discriminant	Accuracy: 83.3%
Last change: Linear Discriminant	3/3 features
2.2 ☆ Quadratic Discriminant	Accuracy: 66.7%
Last change: Quadratic Discriminant	3/3 features
3 ☆ Logistic Regression	Accuracy: 83.3%
Last change: Logistic Regression	3/3 features
4.1 ☆ SVM	Accuracy: 83.3%
Last change: Linear SVM	3/3 features
4.2 ☆ SVM	Accuracy: 91.7%
Last change: Quadratic SVM	3/3 features
4.3 ☆ SVM	Accuracy: 83.3%
Last change: Cubic SVM	3/3 features
4.4 ☆ SVM	Accuracy: 75.0%
Last change: Fine Gaussian SVM	3/3 features
4.5 ☆ SVM	Accuracy: 83.3%
Last change: Medium Gaussian SVM	3/3 features
4.6 ☆ SVM	Accuracy: 58.3%
Last change: Coarse Gaussian SVM	3/3 features
5.1 ☆ KNN	Accuracy: 83.3%
Last change: Fine KNN	3/3 features
5.2 ☆ KNN	Accuracy: 41.7%
Last change: Medium KNN	3/3 features
5.3 ☆ KNN	Accuracy: 41.7%
Last change: Coarse KNN	3/3 features
5.4 ☆ KNN	Accuracy: 41.7%
Last change: Cosine KNN	3/3 features
5.5 ☆ KNN	Accuracy: 41.7%
Last change: Cubic KNN	3/3 features
5.6 ☆ KNN	Accuracy: 83.3%
Last change: Weighted KNN	3/3 features
6.1 ☆ Ensemble	Accuracy: 41.7%
Last change: Boosted Trees	3/3 features
6.2 ☆ Ensemble	Accuracy: 91.7%
Last change: Bagged Trees	3/3 features
6.3 ☆ Ensemble	Accuracy: 83.3%
Last change: Subspace Discriminant	3/3 features
6.4 ☆ Ensemble	Accuracy: 75.0%
Last change: Subspace KNN	3/3 features
6.5 ☆ Ensemble	Accuracy: 41.7%
Last change: RUSBoosted Trees	3/3 features

Figure 3: Overview of the multiple evaluated machine learning pipelines inside the automated framework.

The analysis, classification boundaries, and subsequent plotting structures abide by the following technical definitions:

- **Target Encoding:** A categorical response variable of 1 explicitly represents the ripe state, whereas a value of 0 maps to the unripe state.
- **Scatter Configuration:** Within the generated feature classification spaces, a circular marker (●) indicates a correct prediction, whereas an cross marker (×) represents an incorrect classification.

The MATLAB® Classification Learner application was utilized to train and optimize the diagnostic classifiers using the baseline training set [19]. Three distinct dimensions were established as input features: the relative *MajorAxisLength* ratio, the relative Area ratio, and the core temperature footprint. Principal Component Analysis (PCA) was not executed.

The evaluation benchmark encompassed six primary classification algorithms: DT, discriminant analysis, logistic regression, SVM, *k*-NN, and ensemble classifiers. For each individual algorithm family, the configuration yielding the highest cross-validation prediction accuracy was selected for formal performance comparison, as visualized in Fig. 3.

The optimal structural models were exported from the interactive workspace directly. To evaluate model generalization capability, the exported prediction structures were deployed to process the novel feature profiles from the validation test set via the following inline function execution: `yfit = trainedModel.predictFcn(T);`. This call automatically logs the array of predicted classes into the active environment workspace.

- **Indexing Continuity:** The sample serial numbering system assigned to individual berries was strictly maintained throughout all data logging, processing, and mapping sequences. This index synchronization ensures cross-referencing validity when tracing results between the observation datasets and the final diagnostic feature domains.



Figure 4: Visible spectrum images showing sample indexing for the training set batch (left) and the validation test set batch (right).

3.2.1 Complex DT

A decision tree can be constructed by recursively splitting the source training attributes into uniform subsets based on an optimal homogeneity criterion test [20]. The cross-validation prediction accuracy for the response vector was found to be exceptionally low at 58.3%, yielding an Area Under the Curve (AUC) of 0.66. Graphical performance indicators are presented in Fig. 5.

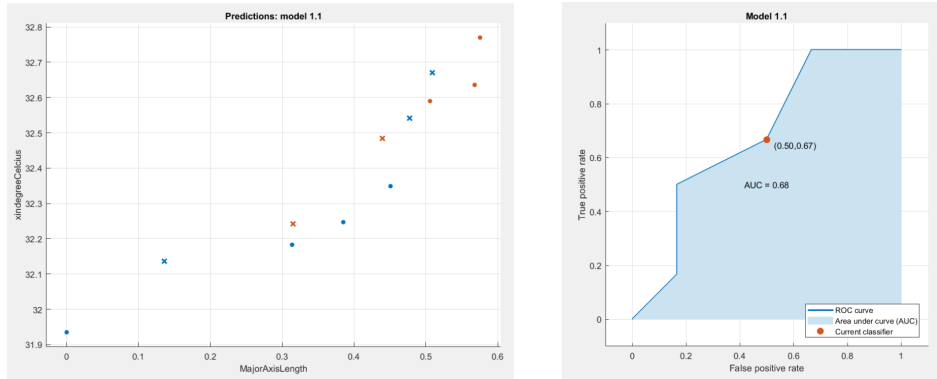


Figure 5: Feature space classification scatter plot (left) and corresponding ROC curve (right) for the Complex DT model.

3.2.2 Quadratic Discriminant Analysis

QDA models assume that numerical observations from individual classes follow a multivariate normal distribution characterized by unique group covariance matrices Σ_y [21]. The underlying classification likelihood ratio is defined mathematically as follows:

$$\text{Likelihood ratio} = \frac{\sqrt{|2\pi\Sigma_{y=1}|^{-1}} \exp\left(-\frac{1}{2}(x - \mu_{y=1})^T \Sigma_{y=1}^{-1}(x - \mu_{y=1})\right)}{\sqrt{|2\pi\Sigma_{y=0}|^{-1}} \exp\left(-\frac{1}{2}(x - \mu_{y=0})^T \Sigma_{y=0}^{-1}(x - \mu_{y=0})\right)} < t \quad (1)$$

The calculated prediction accuracy remained low at 66.7%, with an associated AUC of 0.61, as visualized in Fig. 6.

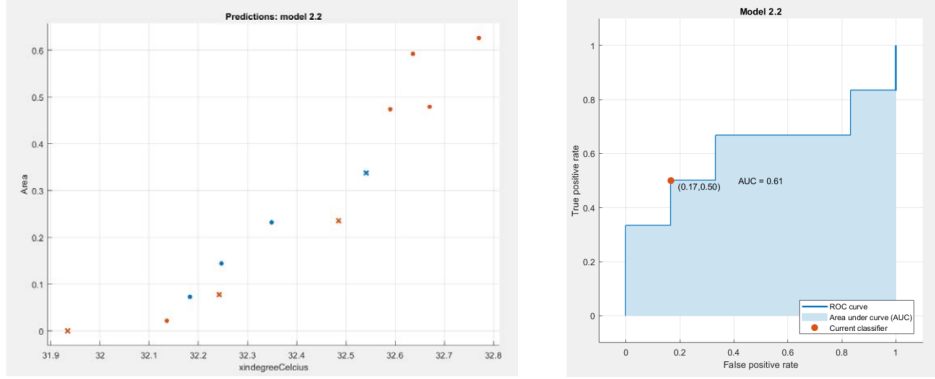


Figure 6: Feature space classification scatter plot (left) and corresponding ROC curve (right) for the QDA model.

3.2.3 Quadratic SVM

The SVM optimization protocol focuses on maximizing the geometric margin given by $\frac{1}{\|W\|}$ subject to the structural linear constraint $W^T X + b \geq 1$. By introducing nonlinear kernel transformations, an optimal quadratic decision boundary profile is generated across the coordinate spaces [19]. This model achieved the highest predictive validation metrics among all standalone supervised classifiers, demonstrating an overall accuracy of 91.7% and an AUC of 0.86, as illustrated in Fig. 7.

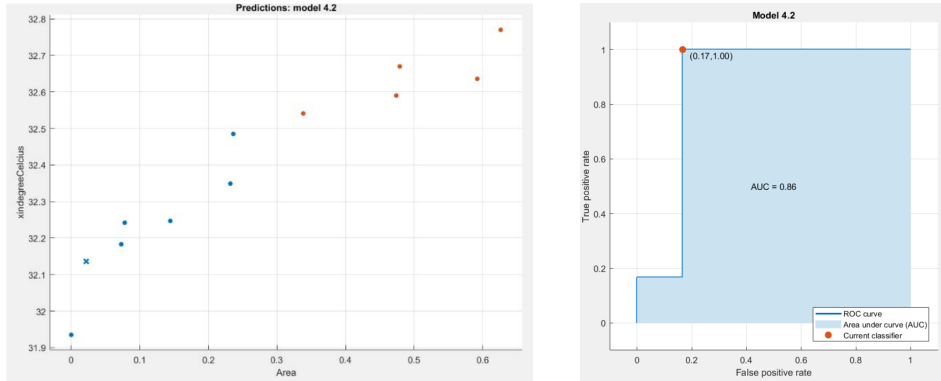


Figure 7: Feature space classification scatter plot (left) and corresponding ROC curve (right) for the Quadratic SVM model.

3.2.4 Fine k -NN Classifier

Within the standard k -NN framework, an unlabelled object vector is assigned to the majority class context calculated among its k nearest spatial neighbors via distance metrics [22]. The Fine k -NN variant restricts localized boundary weights to isolate and eliminate distant side-effect variations, producing a validation accuracy of 83.3% and an AUC of 0.83, as shown in Fig. 8.

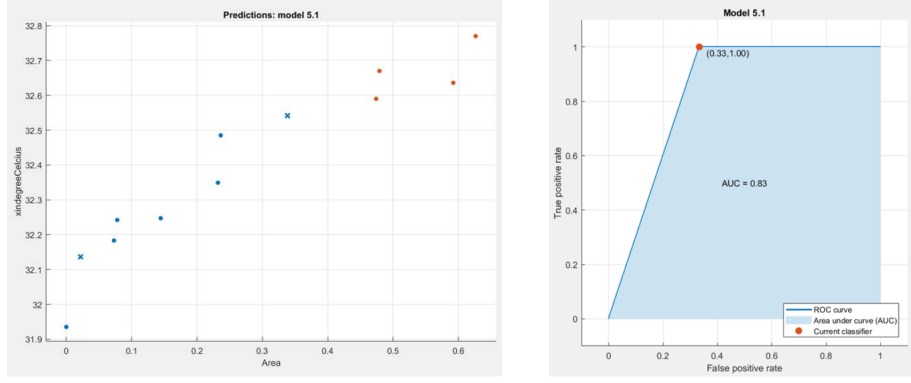


Figure 8: Feature space classification scatter plot (left) and corresponding ROC curve (right) for the Fine k -NN model.

3.2.5 Ensemble Learning: Bagged Trees Architecture

Ensemble methods train multiple diverse hypotheses simultaneously to synthesize optimal collective target decisions [20]. The bootstrap aggregation (bagging) framework significantly reduces variance and mitigates overfitting tendencies across small dataset inputs. This method matched the peak performance threshold, yielding a high validation accuracy of 91.7% and an AUC of 0.86, with comprehensive graphs available in Fig. 9.

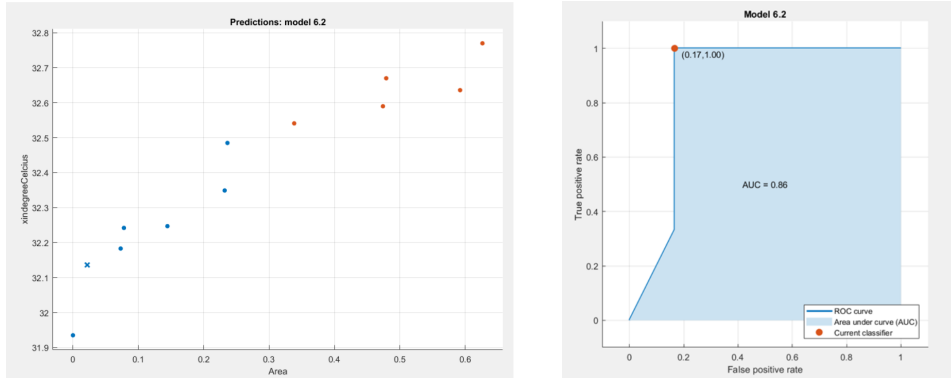


Figure 9: Feature space classification scatter plot (left) and corresponding ROC curve (right) for the Ensemble Bagged Trees model.

3.3 Summary of Performances on Test Batch

The total number of feature sample vectors correctly mapped by each respective classification pipeline during validation is summarized in Table 3. Furthermore, a full architectural breakdown of true versus predicted structural classes is compiled across all optimized models in Table 4.

4 Discussions

Given the limited size of the experimental dataset, selecting classification frameworks capable of maintaining optimal generalization performance was critical. A 2-fold cross-validation scheme is inherently ill-suited for sparse datasets, as it restricts the training matrix to a mere half of the already limited sample pool during each iterative run. Consequently, a leave-one-out strategy or a 5-fold cross-validation configuration represents a more robust evaluation protocol. For this study, a 5-fold cross-validation framework was implemented.

Table 3: Empirical classification tracking results across the independent validation test set.

Berry S.L. No. → Method ↓	Ripe State (Class 1)				Unripe State (Class 0)			Overall Accuracy
	1	2	3	4	1	2	3	
Complex DT	1	1	1	1	0	0	0	100%
Quadratic Disc.	0	0	1	1	0	0	0	71.429%
Quadratic SVM	1	1	0	1	0	0	0	85.714%
Fine k -NN	1	1	1	1	0	0	0	100%
Ensemble (Bagged)	1	1	1	1	0	0	0	100%

Table 4: Confusion matrices compiled for the multi-model berry ripeness binary classification problem.

True Class	Predicted Class											
	Class 0 (Unripe)					Class 1 (Ripe)						
	C.Tree	Quad.	Disc.	QSVM	KNN (fine)	Ensemble	C.Tree	Quad.	Disc.	QSVM	KNN (fine)	Ensemble
Class 0	4	3		6	6	6	2	3		0	0	0
Class 1	3	1		1	2	1	3	5		5	4	5

The choice of classifier architecture plays a decisive role when analyzing small data sets. Among the evaluated methodologies, the standalone decision tree is fundamentally vulnerable to constraints imposed by low-sample data regimes. Due to its greedy splitting characteristics, a small training set frequently precipitates severe overfitting issues. Conversely, the k -nearest neighbors approach can operate effectively on minimalist datasets, requiring a baseline minimum of only k sample vectors, while a standard nearest-neighbor classifier can theoretically execute predictions using a single reference point.

For linear classifiers and SVM models, performance dynamics are heavily governed by the dimensionality of the underlying feature space. SVM architectures are inherently adept at processing sparse datasets because the generation of the optimal separating hyperplane relies exclusively on localized support vectors rather than the global distribution of all data points. Despite this structural advantage, SVM models did not emerge as the single optimal framework in this scenario. A prominent challenge associated with SVM deployments is parameter tuning, an optimization process that becomes highly unstable and difficult to execute reliably when training samples are sparse.

As compiled in Table 4, the validation tracking indicates that the SVM architecture would have demonstrated enhanced testing performance had it been subjected to rigorous, sample-isolated parameter tuning. Furthermore, while the Complex DT model delivered surprisingly adept classifications during the independent testing phase, this outcome is likely a stroke of random statistical luck rather than systemic model reliability, given the greedy nature of its algorithm when restricted to minor data scales.

5 Conclusion

In this work, the thermal signatures of two distinct berry sets were successfully captured via IR spectrum imaging. The baseline dataset served as the calibration set to optimize various supervised machine learning pipelines. The highest-performing model from each core category—DT, discriminant analysis, SVM, k -NN, and ensemble learning—was selected based on its

cross-validation response vector accuracy. During the validation testing phase utilizing an independent test batch, an absolute classification accuracy of 100% was achieved by three out of the five evaluated models. While this flawless accuracy is partly attributable to the small-scale nature of the validation dataset, it firmly demonstrates the viability of non-destructive thermal classification.

Future investigations are required to assess the long-term reliability and environmental resilience of the proposed diagnostic system. Crucially, the transient temperature delta between the inner core and outer fruit surface is sensitive to fluctuations in ambient environmental conditions, which can introduce instability into the model’s prediction boundaries. For instance, washing the berries drastically lowered their baseline thermal readings (reducing both surface and core temperatures by up to 5 °C). This phenomenon was visually apparent when cross-referencing the datasets: certain berry specimens clearly resolved within the visible spectrum imagery in Fig. 4 receded completely from the corresponding IR profiles in Fig. 1 due to the localized cooling effect.

Nevertheless, this study demonstrates that deploying supervised machine learning classifiers configured to analyze relative inner core size and localized core-to-surface temperature gradients provides a highly attractive, computationally efficient, and non-destructive framework for determining the ripeness of commercial agricultural products.

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References

- [1] B. D. Horton. Ripening patterns within a peach as indicated by force and soluble solids concentration. *Journal of the American Society for Horticultural Science*, 117:784–787, 1992.
- [2] M. U. Reyes, R. E. Paull, M. R. Williamson, and L. D. Gautz. Ripeness determination of ‘solo’ papaya (*carica papaya* l.) by impact force. *Applied Engineering in Agriculture*, 12:703–708, 1996.
- [3] W. Kalt and J. E. McDonald. Chemical composition of lowbush blueberry cultivars. *Journal of the American Society for Horticultural Science*, 121:142–146, 1996.
- [4] E. Vangdal. Threshold values of soluble solids in fruit determined for the fresh fruit market. *Acta Agriculturae Scandinavica*, 30:445–448, 1980.
- [5] W. B. McGlasson. Ethylene and fruit ripening. *HortScience*, 18:626, 1983.
- [6] J. R. J. Paré and J. M. R. Bélanger, editors. *Instrumental Techniques in Food Analysis*. Elsevier, Amsterdam, 1997.
- [7] S. I. Cho, G. W. Krutz, H. G. Gibson, and K. Haghighi. Magnet console design of an nmr-based sensor to detect ripeness of fruit. *Transactions of the ASAE*, 33:1043–1050, 1990.
- [8] W. K. Wai, R. L. Stroshine, and G. W. Krutz. Modified hahn echo pulse sequence for proton magnetic resonance (1h-mr) measurements of percent soluble solids of fruits. *Transactions of the ASAE*, 38:849–855, 1995.

- [9] G. Ward and A. Nussinovitch. Peel gloss as a potential indicator of banana ripeness. *Lebensmittel-Wissenschaft und-Technologie*, 29:289–294, 1996.
- [10] N. Galili, I. Shmulevich, and N. Benichou. Acoustic testing of avocado for fruit ripeness evaluation. *Transactions of the ASAE*, 41:399–407, 1998.
- [11] J. L. Sharp. Hot-air quarantine treatment for mangoes infested with caribbean fruit fly. *Journal of Economic Entomology*, 84(4):1262–1265, 1991.
- [12] L. G. Neven. Combined heat and controlled atmosphere quarantine treatments for codling moth in sweet cherries. *Journal of Economic Entomology*, 87(5):1313–1318, 1994.
- [13] S. Lurie. Postharvest heat treatments. *Postharvest Biology and Technology*, 14(3):257–269, 1998.
- [14] H. H. Toba and H. R. Moffitt. Controlled atmosphere and heat combinations for disinfestation of apples. *International Symposium on Postharvest Handling*, 1991.
- [15] L. G. Neven and E. J. Mitcham. Catts (controlled atmosphere temperature treatment system): A non-chemical quarantine treatment for apples and pears. *Postharvest Biology and Technology*, 9, 1996.
- [16] K. C. Shellie and R. L. Mangan. Postharvest heat treatments for grapefruit infusion control. *HortScience*, 1997.
- [17] J. D. Hansen. Heating times for fruit center temperature under hot-water immersion and forced hot-air treatments. *Postharvest Biology and Technology*, 1992.
- [18] S. Wang, J. Tang, and R. P. Cavalieri. Modeling fruit internal heating rates for hot air and hot water treatments. *Postharvest Biology and Technology*, 22(3):257–270, 2001.
- [19] Issam Dagher. Quadratic kernel support vector machine. *Journal of Global Optimization*, 41(1):15–30, 2008.
- [20] J. Ross Quinlan. Induction of decision trees. *Machine Learning*, 1(1):81–106, 1986.
- [21] Thomas M. Cover. Geometrical and statistical properties of systems of linear inequalities with applications in pattern recognition. *IEEE Transactions on Electronic Computers*, EC-14(3):326–334, 1965.
- [22] Naomi S. Altman. An introduction to kernel and nearest-neighbor nonparametric regression. *The American Statistician*, 46(3):175–185, 1992.